Netflix Data Science Project:

Introduction

This notebook performs a comprehensive Exploratory Data Analysis (EDA) on a Netflix dataset. The goal is to uncover trends in content creation, understand distribution across countries, and categorize content for better insights.

Steps covered:

1. **Data Cleaning:** Handling missing values and duplicates.
2. **Feature Engineering:** Creating new metrics like `year_added`, `cast_count`, and `rating_category`.
3. **Exploratory Analysis:** Statistical summaries and visual inspections.

In [ ]:

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```

1. Environment Setup and Data Loading

In this section, we import the necessary libraries for data manipulation (`pandas`, `numpy`) and visualization (`seaborn`, `matplotlib`). We also load the raw Netflix dataset.

In [105]:

```
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt
import numpy as np

from warnings import filterwarnings
filterwarnings("ignore")



df = pd.read_csv("dataset/Netflix.csv")
```

2. Initial Data Inspection

Before cleaning, we take a quick look at the first few rows of the data, its dimensions, and the data types of each column to identify potential issues.

In [106]:

```
df.head()
```

Out[106]:

	show_id	type	title	director	cast	country	date_added	release_year	rating	duration	genres
0	s1	TV Show	3%	NaN	João Miguel, Bianca Comparato, Michel Gomes, R...	Brazil	14-Aug-20	2020	TV-MA	4	International TV Shows, TV Dramas, TV Sci-Fi &...
1	s10	Movie	1920	Vikram Bhatt	Rajneesh Duggal, Adah Sharma, Indraneil Sengup...	India	15-Dec-17	2008	TV-MA	143	Horror Movies, International Movies, Thrillers
2	s100	Movie	3 Heroines	Iman Brotoseno	Reza Rahadian, Bunga Citra Lestari, Tara Basro...	Indonesia	5-Jan-19	2016	TV-PG	124	Dramas, International Movies, Sports Movies
3	s1000	Movie	Blue Mountain State: The Rise of Thadland	Lev L. Spiro	Alan Ritchson, Darin Brooks, James Cade, Rob R...	United States	1-Mar-16	2016	R	90	Comedies
4	s1001	TV Show	Blue Planet II	NaN	David Attenborough	United Kingdom	3-Dec-18	2017	TV-G	1	British TV Shows, Docuseries, Science & Nature TV

In [107]:

```
df.shape
```

Out[107]:

```
(7787, 12)
```

In [108]:

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 7787 entries, 0 to 7786
Data columns (total 12 columns):
 #   Column        Non-Null Count  Dtype 
---  ------        --------------  ----- 
 0   show_id       7787 non-null   object
 1   type          7787 non-null   object
 2   title         7787 non-null   object
 3   director      5398 non-null   object
 4   cast          7069 non-null   object
 5   country       7280 non-null   object
 6   date_added    7777 non-null   object
 7   release_year  7787 non-null   int64 
 8   rating        7780 non-null   object
 9   duration      7787 non-null   int64 
 10  genres        7787 non-null   object
 11  description   7787 non-null   object
dtypes: int64(2), object(10)
memory usage: 730.2+ KB
```

In [109]:

```
df.describe()
```

Out[109]:

	release_year	duration
count	7787.000000	7787.000000
mean	2013.932580	69.122769
std	8.757395	50.950743
min	1925.000000	1.000000
25%	2013.000000	2.000000
50%	2017.000000	88.000000
75%	2018.000000	106.000000
max	2021.000000	312.000000

In [110]:

```
df.isnull().sum()
```

Out[110]:

```
show_id            0
type               0
title              0
director        2389
cast             718
country          507
date_added        10
release_year       0
rating             7
duration           0
genres             0
description        0
dtype: int64
```

In [111]:

```
df.duplicated().sum()
```

Out[111]:

```
np.int64(0)
```

3. Data Cleaning

To ensure the quality of our analysis, we perform the following:

- **Drop Unnecessary Columns:** Removing `show_id` as it serves only as a unique identifier.
- **Impute Missing Values:** Filling gaps in `director`, `cast`, and `country` with "Unknown", and using the mode for `rating` and `date_added`.

In [112]:

```
df.drop(columns=['show_id'], inplace=True)
```

In [113]:

```
df['director'].fillna('Unknown', inplace=True)
df['cast'].fillna('Unknown', inplace=True)
df['country'].fillna('Unknown', inplace=True)
df['rating'].fillna(df['rating'].mode()[0], inplace=True)
df['date_added'].fillna(df['date_added'].mode()[0], inplace=True)
```

4. Feature Engineering

We create new features to extract more value from the existing data:

1. **Date Extraction:** Converting `date_added` to datetime objects to extract year and month.
2. **Complexity Metrics:** Calculating the number of cast members and genres per title.
3. **Categorization:** Simplifying the content ratings into broader categories: **Adult**, **Teen**, and **Kids**.
4. **Localization:** Extracting the primary country of origin from the `country` column.

In [114]:

```
df['date_added'] = pd.to_datetime(df['date_added'])

df['year_added'] = df['date_added'].dt.year
df['month_added'] = df['date_added'].dt.month
```

In [115]:

```
df.drop(columns=['date_added'], inplace=True)
```

In [116]:

```
df['cast_count'] = df['cast'].apply(lambda x: len(x.split(',')))
df['genre_count'] = df['genres'].apply(lambda x: len(x.split(',')))
```

In [117]:

```
df['duration_type'] = df['type']

df.rename(columns={'duration':'duration_value'}, inplace=True)
```

In [118]:

```
df['country_main'] = df['country'].apply(lambda x: x.split(',')[0])
```

In [119]:

```
def simplify_rating(r):
    if r in ['TV-MA','R','NC-17']:
        return "Adult"
    elif r in ['PG-13','TV-14']:
        return "Teen"
    else:
        return "Kids"

df['rating_category'] = df['rating'].apply(simplify_rating)
```

In [120]:

```
df.drop(columns=['rating'], inplace=True)
```

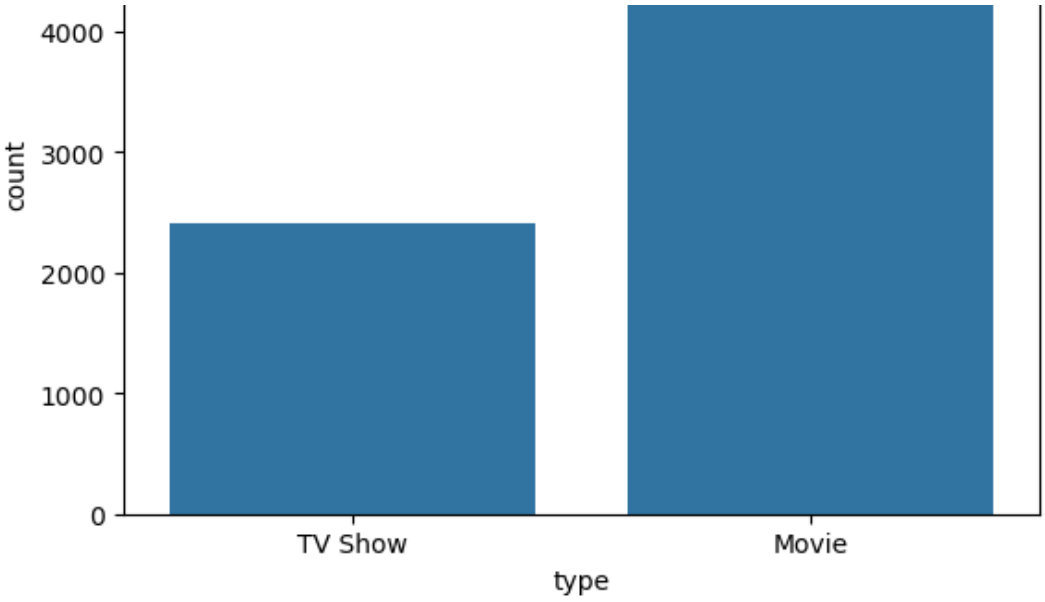
5. Exploratory Data Analysis & Visualization

With the data cleaned and enriched, we now move to the visualization phase. This help us identify:

- **The distribution of content types (Movies vs. TV Shows).**
- **Trends in content releases over the years.**
- **Relationship between content duration and its category.**

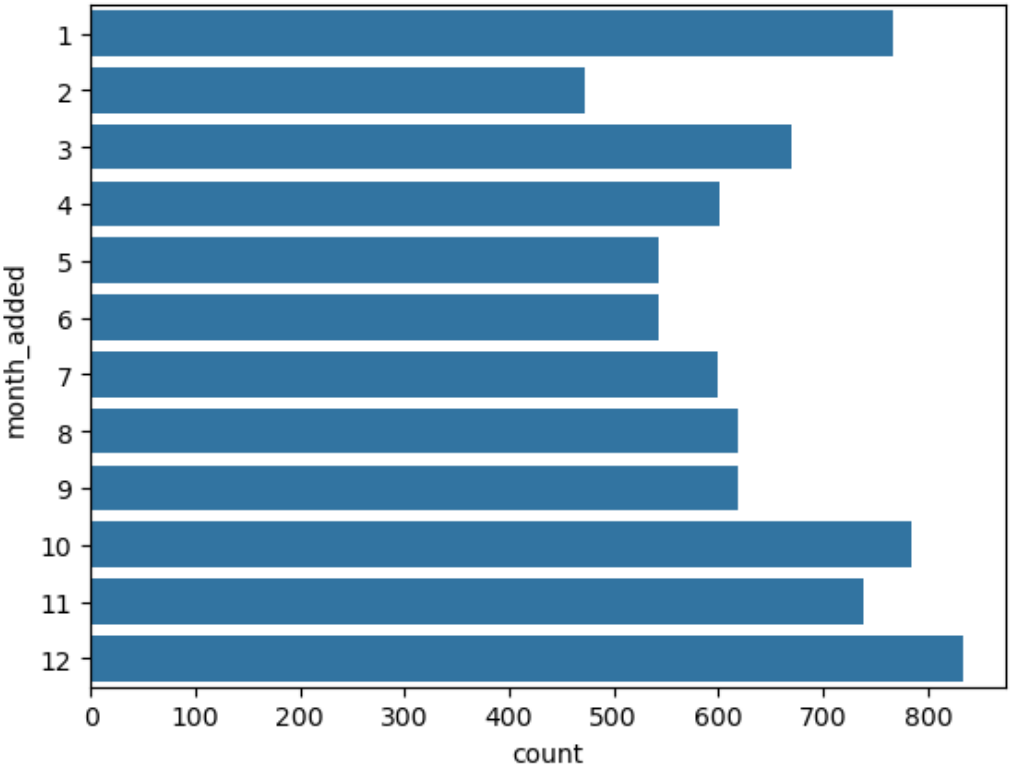
In [121]:

```
sns.countplot(x="type",data=df)
plt.show()
```



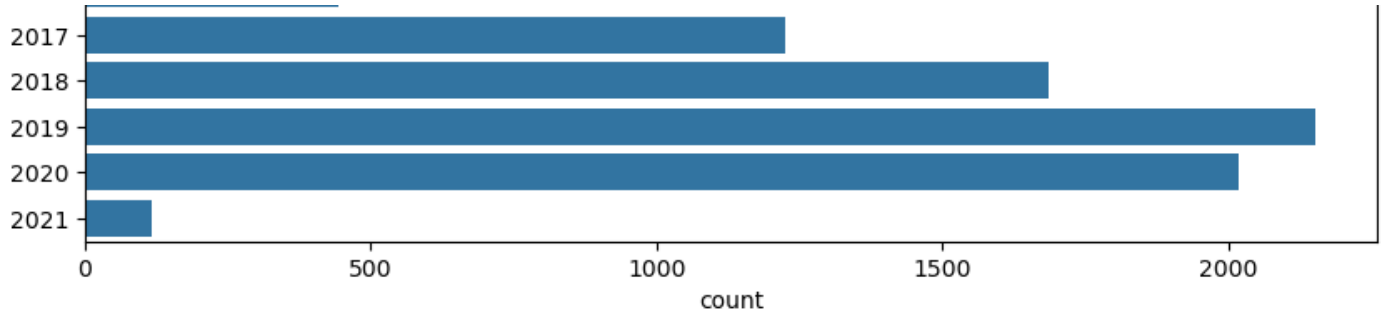
In [122]:

```
sns.countplot(y="month_added",data=df)
plt.show()
```



In [123]:

```
plt.figure(figsize=(10,5))
sns.countplot(y=df["year_added"])
plt.show()
```



In [ ]:

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In [ ]:

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```

In [124]:

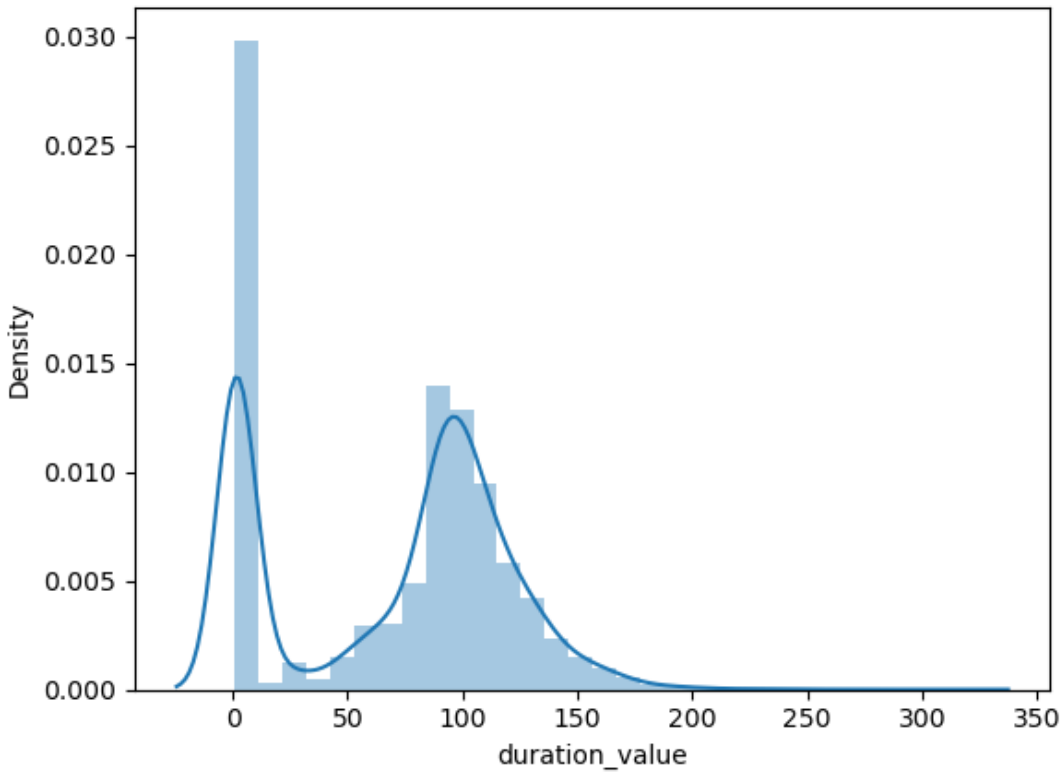
```
df['has_director'] = df['director'].apply(lambda x: 0 if x=='Unknown' else 1)
```

In [125]:

```
sns.distplot(df["duration_value"])
```

Out[125]:

```
<Axes: xlabel='duration_value', ylabel='Density'>
```



In [126]:

```
df.corr(numeric_only=True)["duration_value"].reset_index()
```

Out[126]:

	index	duration_value
0	release_year	-0.240836
1	duration_value	1.000000
2	year_added	0.037102
3	month_added	-0.024913

	index	duration_value
4	cast_count	0.096601
5	genre_count	0.092256
6	has_director	0.820335

In [127]:

```
df.corr(numeric_only=True)
```

Out[127]:

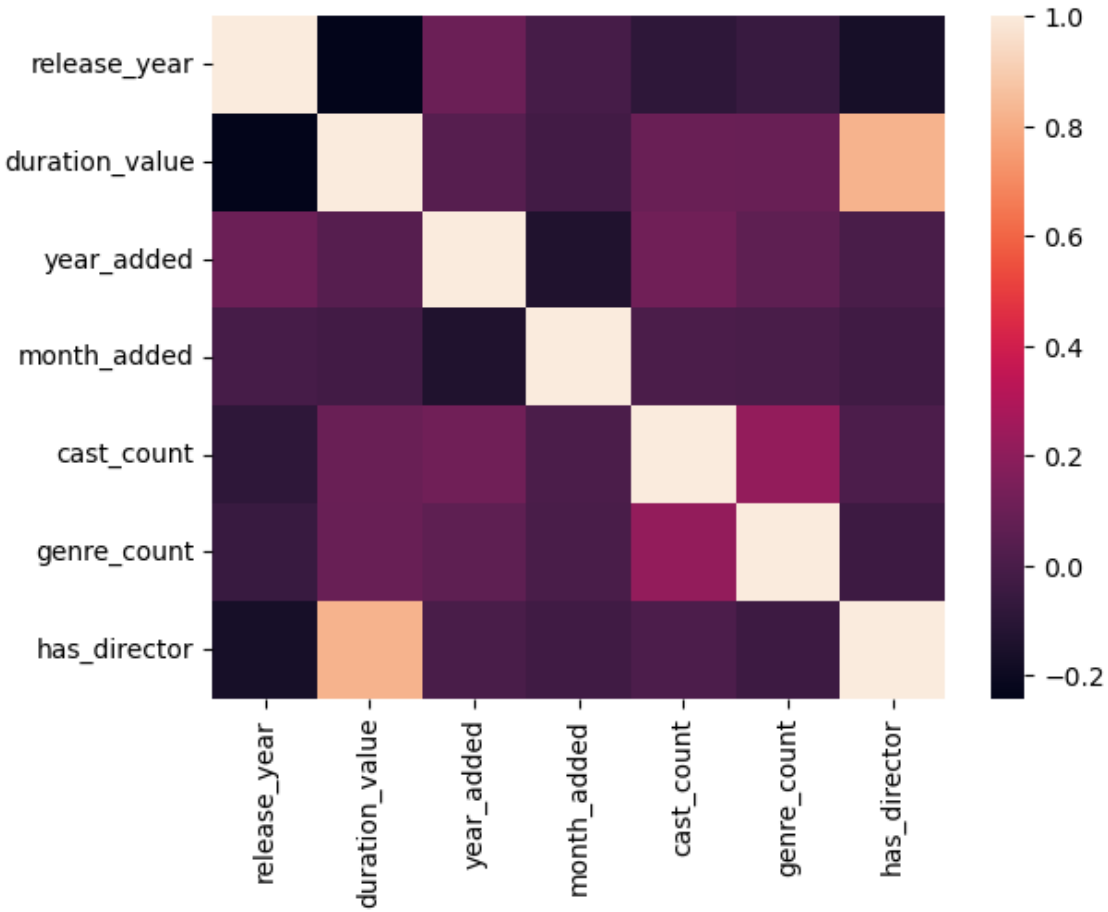
	release_year	duration_value	year_added	month_added	cast_count	genre_count	has_director
release_year	1.000000	-0.240836	0.100115	-0.009661	-0.086585	-0.052931	-0.166603
duration_value	-0.240836	1.000000	0.037102	-0.024913	0.096601	0.092256	0.820335
year_added	0.100115	0.037102	1.000000	-0.135006	0.113470	0.060552	-0.001745
month_added	-0.009661	-0.024913	-0.135006	1.000000	0.001902	-0.002332	-0.028360
cast_count	-0.086585	0.096601	0.113470	0.001902	1.000000	0.219182	0.008707
genre_count	-0.052931	0.092256	0.060552	-0.002332	0.219182	1.000000	-0.041000
has_director	-0.166603	0.820335	-0.001745	-0.028360	0.008707	-0.041000	1.000000

In [128]:

```
sns.heatmap(df.corr(numeric_only=True))
```

Out[128]:

```
<Axes: >
```



Q1] CONTENT-TYPE LABELER:

In [129]:

```
df.head()
```

Out[129]:

	type	title	director	cast	country	release_year	duration_value	genres	description	year_added
0	TV Show	3%	Unknown	João Miguel, Bianca Comparato, Michel Gomes, R...	Brazil	2020	4	International TV Shows, TV Dramas, TV Sci-Fi &...	In a future where the elite inhabit an island ...	2020
1	Movie	1920	Vikram Bhatt	Rajneesh Duggal, Adah Sharma, Indraneil Sengup...	India	2008	143	Horror Movies, International Movies, Thrillers	An architect and his wife move into a castle t...	2017
2	Movie	3 Heroines	Iman Brotoseno	Reza Rahadian, Bunga Citra Lestari, Tara Basro...	Indonesia	2016	124	Dramas, International Movies, Sports Movies	Three Indonesian women break records by becomi...	2019
3	Movie	Blue Mountain State: The Rise of Thadland	Lev L. Spiro	Alan Ritchson, Darin Brooks, James Cade, Rob R...	United States	2016	90	Comedies	New NFL star Thad buys his old teammates' belo...	2016
4	TV Show	Blue Planet II	Unknown	David Attenborough	United Kingdom	2017	1	British TV Shows, Docuseries, Science & Nature TV	This sequel to the award-winning nature series...	2018

In [ ]:

In [ ]:

LOGISTIC - REGRESSION:

In [130]:

```
# Import necessary libraries
from sklearn.model_selection import train_test_split
from sklearn.compose import ColumnTransformer
from sklearn.preprocessing import OneHotEncoder, StandardScaler
from sklearn.linear_model import LogisticRegression
from sklearn.pipeline import Pipeline
from sklearn.metrics import accuracy_score, classification_report, confusion_matrix
import pickle


num_features = ['cast_count', 'genre_count', 'duration_value']

cat_features = ['country_main', 'rating_category']

X = df[num_features + cat_features]
y = df['type']

# Split the data into train and test sets (80-20 split)
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42
, stratify=y)
```

```python
num_transformer = Pipeline(steps=[
    ('scaler', StandardScaler())
])


cat_transformer = Pipeline(steps=[
    ('onehot', OneHotEncoder(drop='first', handle_unknown='ignore'))
])


preprocessor = ColumnTransformer(
    transformers=[
        ('num', num_transformer, num_features),
        ('cat', cat_transformer, cat_features)
    ])



prediction_pipeline_1 = Pipeline(steps=[
    ('preprocessor', preprocessor),
    ('classifier', LogisticRegression(random_state=42, max_iter=1000))
])


prediction_pipeline_1.fit(X_train, y_train)


y_pred = prediction_pipeline_1.predict(X_test)

# Evaluate the model
accuracy = accuracy_score(y_test, y_pred)
print(f"Accuracy: {accuracy:.4f}")
print("\nClassification Report:")
print(classification_report(y_test, y_pred, target_names=['Movie', 'TV Show']))
print("\nConfusion Matrix:")
print(confusion_matrix(y_test, y_pred))

# Save the pipeline model (similar to laptop project)
with open('netflix_type_predictor_pipeline_1.pkl', 'wb') as f:
    pickle.dump(prediction_pipeline_1, f)


# Example prediction function (like app.py in laptop project)
def predict_content_type(cast_count, genre_count, duration_value,
                         country_main, rating_category):
    # Create input DataFrame
    query_df = pd.DataFrame({
        'cast_count': [cast_count],
        'genre_count': [genre_count],
        'duration_value': [duration_value],
        'country_main': [country_main],
        'rating_category': [rating_category]
    })

    # Load the pipeline
    with open('netflix_type_predictor_pipeline_1.pkl', 'rb') as f:
        pipe = pickle.load(f)

    # Predict
    prediction = pipe.predict(query_df)[0]
    content_type = 'TV Show' if prediction == 1 else 'Movie'
    probability = pipe.predict_proba(query_df)[0]
    conf_movie = probability[0] * 100
    conf_tv = probability[1] * 100

    return f"Predicted Type: {content_type} (Confidence: {max(conf_movie, conf_tv):.2f}%)
"

# Example usage
print("\nExample Prediction:")
```

```
result = predict_content_type(cast_count=8, genre_count=3, duration_value=143,
    country_main='India', rating_category='Adult'
)
print(result)
```

```
Accuracy: 0.9961

Classification Report:
              precision    recall  f1-score   support

       Movie       1.00      0.99      1.00      1076
     TV Show       0.99      1.00      0.99       482

    accuracy                           1.00      1558
   macro avg       0.99      1.00      1.00      1558
weighted avg       1.00      1.00      1.00      1558


Confusion Matrix:
[[1070    6]
 [   0  482]]

Example Prediction:
Predicted Type: Movie (Confidence: 100.00%)
```

In [ ]:

NAIVE - BAYES (Bernoulli) :

In [131]:

```
from sklearn.naive_bayes import BernoulliNB

prediction_pipeline_2 = Pipeline(steps=[
    ('preprocessor', preprocessor),
    ('classifier', BernoulliNB(
    alpha=1.0,
    binarize=0.0,
    fit_prior=True,
    class_prior=None
))
])


prediction_pipeline_2.fit(X_train, y_train)


y_pred = prediction_pipeline_2.predict(X_test)

# Evaluate the model
accuracy = accuracy_score(y_test, y_pred)
print(f"Accuracy: {accuracy:.4f}")
print("\nClassification Report:")
print(classification_report(y_test, y_pred, target_names=['Movie', 'TV Show']))
print("\nConfusion Matrix:")
print(confusion_matrix(y_test, y_pred))

# Save the pipeline model (similar to laptop project)
with open('netflix_type_predictor_pipeline_2.pkl', 'wb') as f:
    pickle.dump(prediction_pipeline_2, f)
```

```
Accuracy: 0.8954

Classification Report:
              precision    recall  f1-score   support

       Movie       0.98      0.86      0.92      1076
     TV Show       0.76      0.96      0.85       482
```

```
    accuracy                           0.90      1558
   macro avg       0.87      0.91      0.89      1558
weighted avg       0.91      0.90      0.90      1558


Confusion Matrix:
[[930 146]
 [ 17 465]]
```

In [ ]:

```

```

KNN:

In [132]:

```
from sklearn.neighbors import KNeighborsClassifier

prediction_pipeline_3 = Pipeline(steps=[
    ('preprocessor', preprocessor),
    ('classifier', KNeighborsClassifier(
        n_neighbors=5,
        weights='uniform',
        algorithm='auto',
        metric='minkowski',
        p=2
    ))
])

prediction_pipeline_3.fit(X_train, y_train)

y_pred = prediction_pipeline_3.predict(X_test)

# Evaluate the model
accuracy = accuracy_score(y_test, y_pred)
print(f"Accuracy: {accuracy:.4f}")

print("\nClassification Report:")
print(classification_report(y_test, y_pred, target_names=['Movie', 'TV Show']))

print("\nConfusion Matrix:")
print(confusion_matrix(y_test, y_pred))

# Save the pipeline model
with open('netflix_type_predictor_knn_pipeline_3.pkl', 'wb') as f:
    pickle.dump(prediction_pipeline_3, f)
```

```
Accuracy: 0.9891

Classification Report:
              precision    recall  f1-score   support

       Movie       1.00      0.99      0.99      1076
     TV Show       0.97      1.00      0.98       482

    accuracy                           0.99      1558
   macro avg       0.98      0.99      0.99      1558
weighted avg       0.99      0.99      0.99      1558


Confusion Matrix:
[[1060   16]
 [   1  481]]
```

In [ ]:

```

```

In [ ]:

In [ ]:

In [ ]:

In [ ]:

CONCLUSION: Highest accuracy is acheived through Logistic Regression.

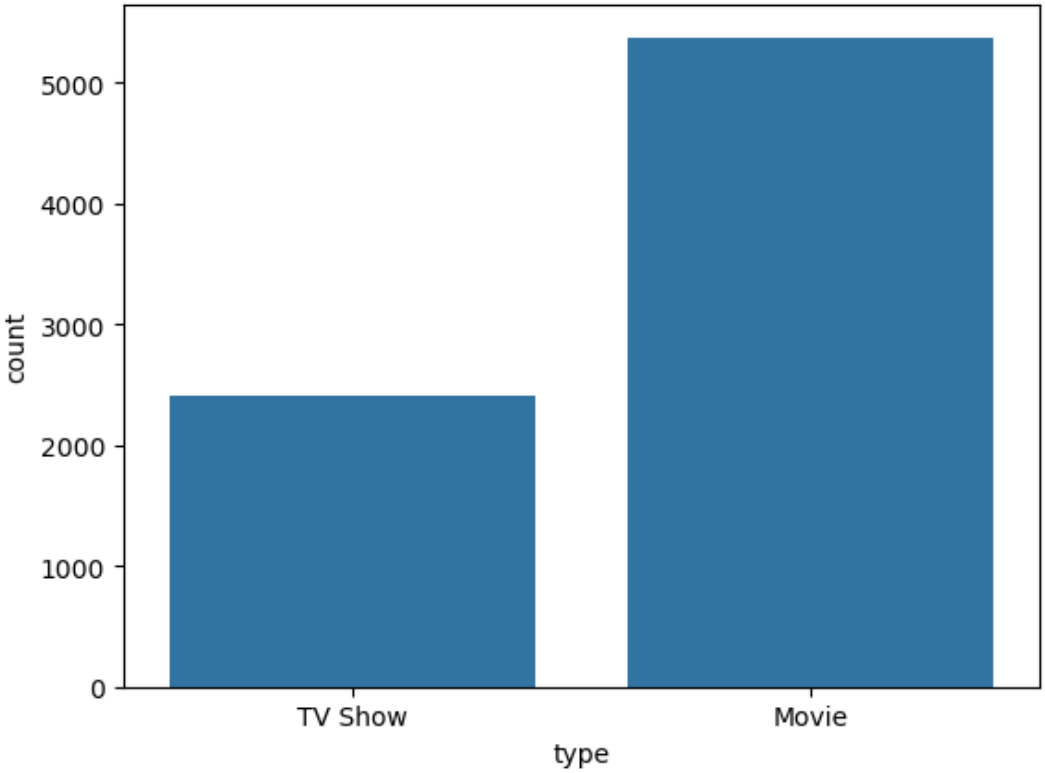
In [ ]:

In [ ]:

Q2] Predict whether a piece of content is a Movie or a TV Show based on features like cast-size, country, and release trends?

In [133]:

```
sns.countplot(x=df["type"])
plt.show()
```



In [134]:

```
data = df.groupby("type")["genre_count"].sum().reset_index()
data
```

Out[134]:

	type	genre_count
0	Movie	11546

	type	genre_count
1	TV Show	9925

In [135]:

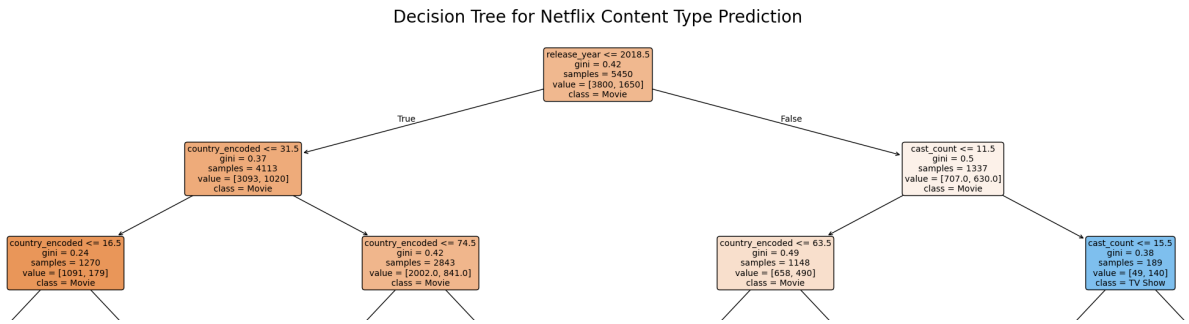
```
data = df.groupby("type")["cast_count"].sum().reset_index()
data
```

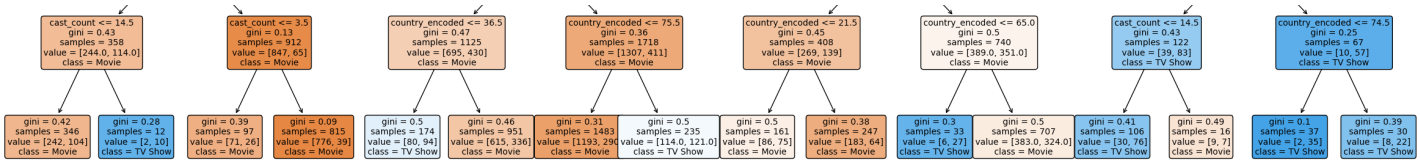
Out[135]:

	type	cast_count
0	Movie	38738
1	TV Show	17935

In [136]:

```
from sklearn.model_selection import train_test_split
from sklearn.tree import DecisionTreeClassifier, plot_tree
import matplotlib.pyplot as plt
from sklearn.preprocessing import LabelEncoder

# 1. Preprocessing categorical features
# We extract the first country and first genre to simplify the categories
df['first_country'] = df['country'].fillna('Unknown').apply (lambda x: x.split(',')[0])

# 2. Encode strings to numbers
le_country = LabelEncoder()

df['country_encoded'] = le_country.fit_transform(df['first_country'])



feature_cols = ["cast_count", "country_encoded", "release_year"]
X = df[feature_cols]
y = df["type"]


X_train , X_test , y_train, y_test = train_test_split(X,y, test_size=0.3,random_state=42
)


model = DecisionTreeClassifier(max_depth=4, random_state=42)
model.fit(X_train , y_train)


plt.figure(figsize=(30,10))
plot_tree(model,
          feature_names=feature_cols,
          class_names=["Movie", "TV Show"],
          filled=True,
          rounded=True,
          fontsize=10,
          precision=2)

plt.title("Decision Tree for Netflix Content Type Prediction", fontsize=20)
plt.show()


print("Accuracy:",model.score(X_test, y_test))
```





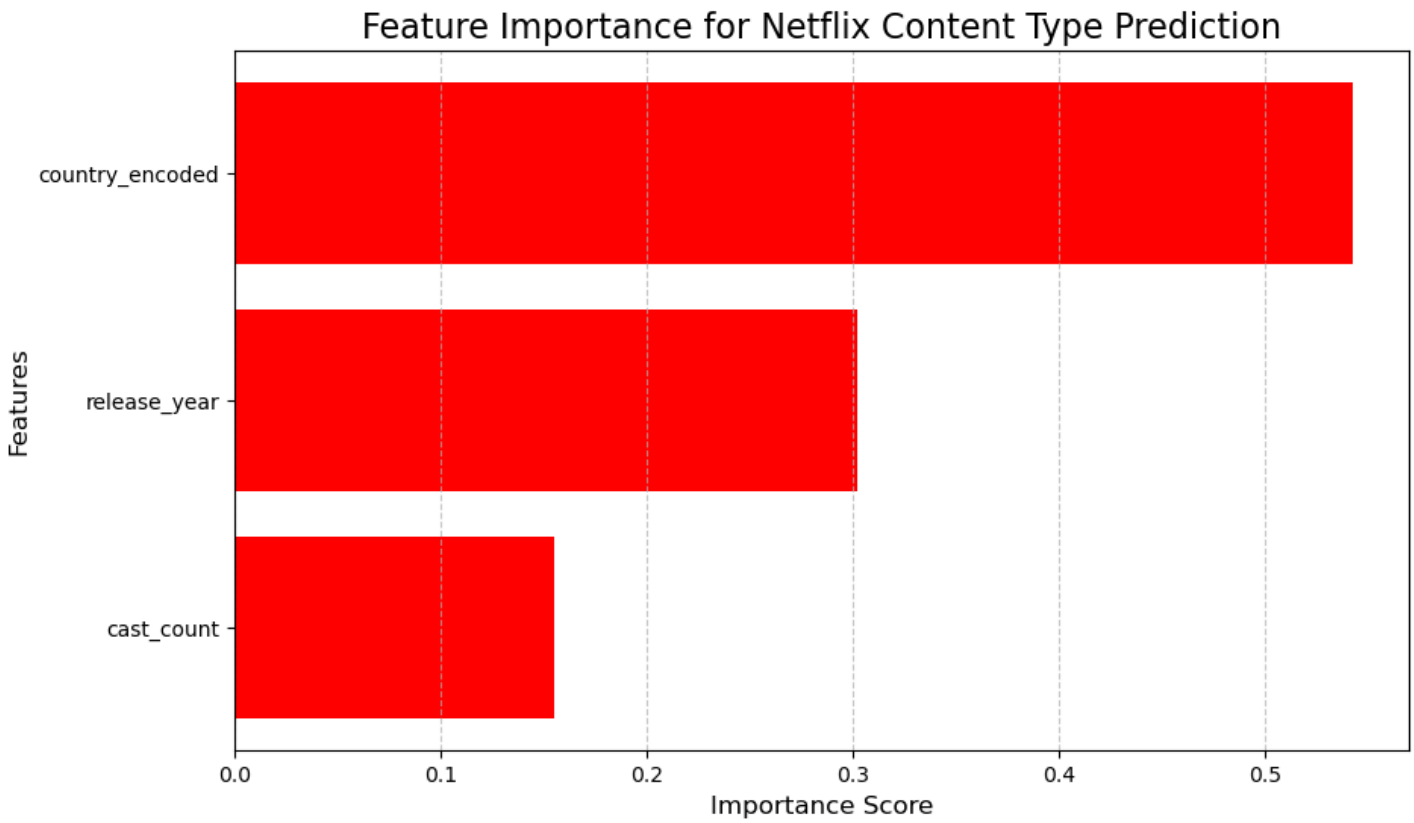

```
Accuracy: 0.7064612751390672
```

In [ ]:

```

```

In [137]:

```
# 1. Create a DataFrame to hold the feature names and their corresponding importance scor
es
feature_importance_df = pd.DataFrame({
    'Feature': feature_cols,
    'Importance': model.feature_importances_
}).sort_values(by='Importance', ascending=True)

# 2. Plot the Feature Importance
plt.figure(figsize=(10, 6))
plt.barh(feature_importance_df['Feature'], feature_importance_df['Importance'], color='re
d')

plt.title('Feature Importance for Netflix Content Type Prediction', fontsize=16)
plt.xlabel('Importance Score', fontsize=12)
plt.ylabel('Features', fontsize=12)
plt.grid(axis='x', linestyle='--', alpha=0.7)

plt.show()

# 3. Print the values for reference
# Create the sorted dataframe and drop the old index
importance_display = feature_importance_df.sort_values(by='Importance', ascending=False)
.reset_index(drop=True)

# Display the clean table
importance_display.reset_index()
```



Out[137]:

	index	Feature	Importance
0	0	country_encoded	0.542345
1	1	release_year	0.302328
2	2	cast_count	0.155326

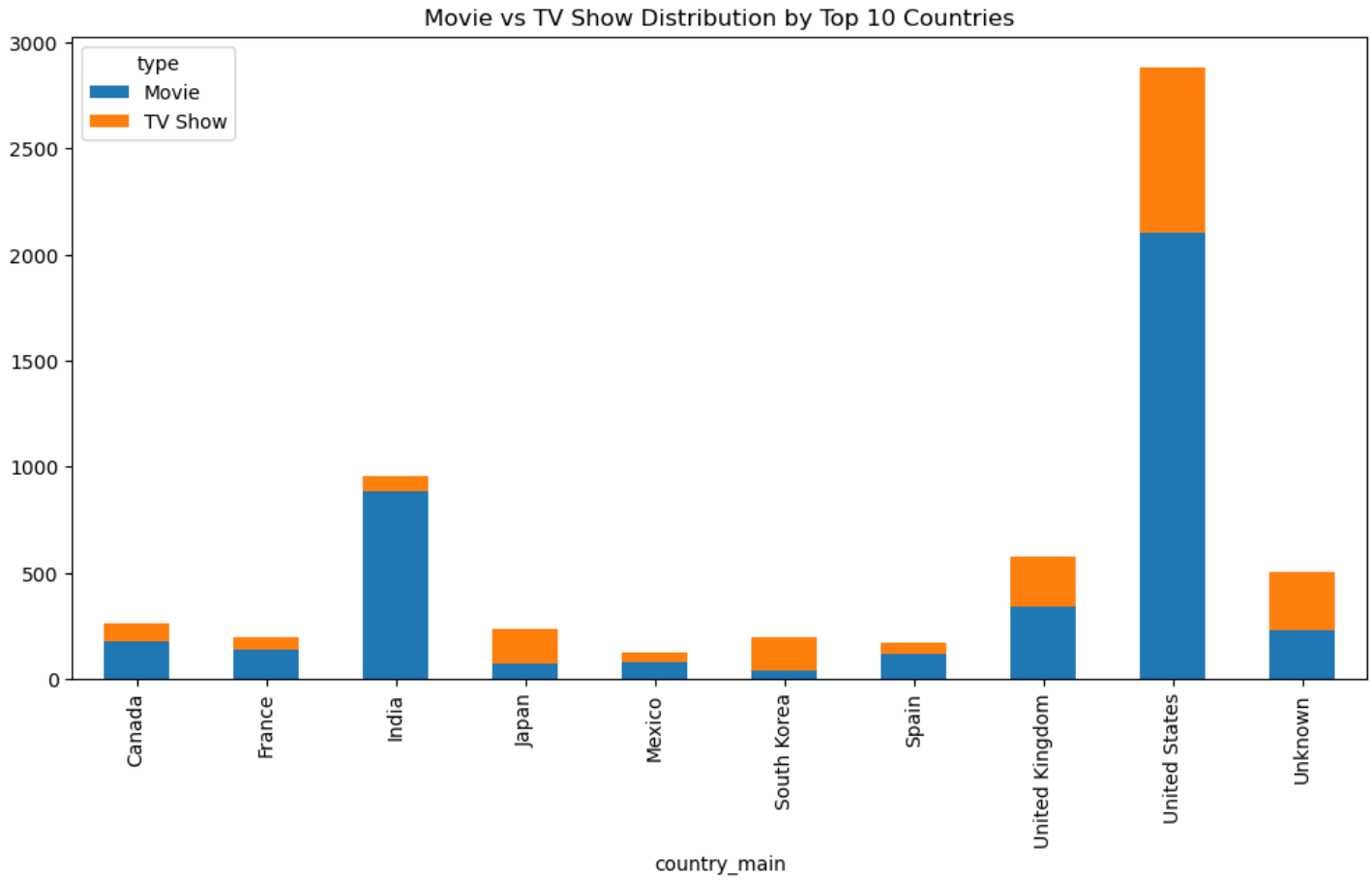
In [ ]:

```

```

Q3] Which countries dominate Netflix content, and how does the Movie vs TV-Show ratio change across the top 10 countries?

In [138]:

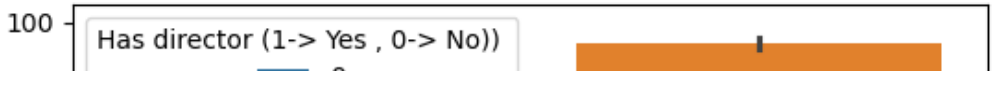
```
top10_countries = df['country_main'].value_counts().head(10).index
country_type = df[df['country_main'].isin(top10_countries)].groupby(['country_main', 'ty
pe']).size().unstack()
country_type.plot(kind='bar', stacked=True, figsize=(12,6))
plt.title('Movie vs TV Show Distribution by Top 10 Countries')
plt.show()
```

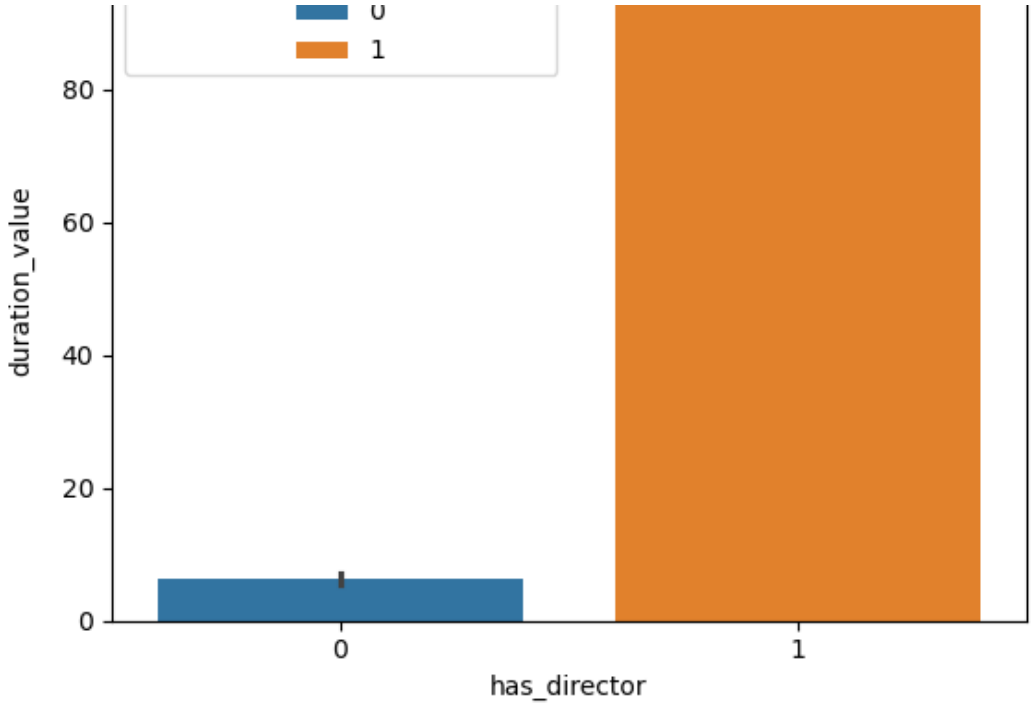


Q4] Do titles with a known director have significantly higher cast_count, genre_count, or longer duration than “Unknown” director titles?

In [139]:

```
df.groupby('has_director')[['cast_count', 'genre_count', 'duration_value']].mean().round
(2)
sns.barplot(x='has_director', y='duration_value',hue="has_director", data=df)
plt.legend(title="Has director (1-> Yes , 0-> No)) ")
plt.show()
```





In [140]:

```
df.head()
```

Out[140]:

	type	title	director	cast	country	release_year	duration_value	genres	description	year_added
0	TV Show	3%	Unknown	João Miguel, Bianca Comparato, Michel Gomes, R...	Brazil	2020	4	International TV Shows, TV Dramas, TV Sci-Fi &...	In a future where the elite inhabit an island ...	2020
1	Movie	1920	Vikram Bhatt	Rajneesh Duggal, Adah Sharma, Indraneil Sengup...	India	2008	143	Horror Movies, International Movies, Thrillers	An architect and his wife move into a castle t...	2017
2	Movie	3 Heroines	Iman Brotoseno	Reza Rahadian, Bunga Citra Lestari, Tara Basro...	Indonesia	2016	124	Dramas, International Movies, Sports Movies	Three Indonesian women break records by becomi...	2019
3	Movie	Blue Mountain State: The Rise of Thadland	Lev L. Spiro	Alan Ritchson, Darin Brooks, James Cade, Rob R...	United States	2016	90	Comedies	New NFL star Thad buys his old teammates' belo...	2016
4	TV Show	Blue Planet II	Unknown	David Attenborough	United Kingdom	2017	1	British TV Shows, Docuseries, Science & Nature TV	This sequel to the award-winning nature series...	2018

CONTENT- RECOMMENDATION SYSTEM:

In [141]:

```
import pandas as pd
import pickle
```

```python
from nltk.stem.porter import PorterStemmer
from sklearn.feature_extraction.text import TfidfVectorizer
from sklearn.metrics.pairwise import cosine_similarity

ps = PorterStemmer()

def stem(text):
    y = [ps.stem(i) for i in str(text).split()]
    return " ".join(y)

df['tags'] = (
    df['title'].fillna('') + " " +
    df['genres'].fillna('') + " " +
    df['description'].fillna('') + " " +
    df['cast'].fillna('') + " " +
    df['director'].fillna('') + " " +
    df['country_main'].fillna('') + " " +
    df['rating_category'].fillna('') + " " +
    df['type'].fillna('')
)

df['tags'] = df['tags'].apply(lambda x: x.lower())
df['tags'] = df['tags'].apply(stem)

tfidf = TfidfVectorizer(max_features=10000, stop_words='english')
vectors = tfidf.fit_transform(df['tags']).toarray()


class RecommendationModel:
    def __init__(self, vectorizer, vectors, df):
        self.vectorizer = vectorizer
        self.vectors = vectors
        self.df = df.reset_index(drop=True)

    def recommend(self, content, top_n=10):
        mask = self.df['title'].str.contains(content, case=False, na=False)
        if mask.any():
            idx = self.df[mask].index[0]
            query_vec = self.vectors[idx].reshape(1, -1)
            print(f" Found exact title → using full features")
        else:
            query_vec = self.vectorizer.transform([stem(content.lower())]).toarray()

        similarity = cosine_similarity(query_vec, self.vectors)[0]

        content_list = sorted(
            list(enumerate(similarity)),
            reverse=True,
            key=lambda x: x[1]
        )[1:top_n+1]

        print(f"\n\nRecommendations similar to: **{content}** \n")
        for rank, (i, score) in enumerate(content_list, 1):
            row = self.df.iloc[i]
            print(f"{rank:2}) • {row['title']}  "
                  f"         |  {row['type']}  "
                  f"         |  {row['genres'][:70]}...  ")


model = RecommendationModel(tfidf, vectors, df)

with open("content_recommendation_model_improved.pkl", "wb") as f:
    pickle.dump(model, f)


with open("content_recommendation_model_improved.pkl", "rb") as f:
    loaded_model = pickle.load(f)

loaded_model.recommend("Kesari 2")
loaded_model.recommend("Dune")
loaded_model.recommend("Stranger Things")
loaded_model.recommend("Article 15")
```

```
loaded_model.recommend("Dhurandhar")
loaded_model.recommend("Dhurandhar - The Revenge")
loaded_model.recommend("Dear Zindagi")
```

```
Recommendations similar to: **Kesari 2**

 1) • 1920            |  Movie            |  Horror Movies, International Movies, Thriller
s...
 2) • 3 Heroines            |  Movie            |  Dramas, International Movies, Sports Mo
vies...
 3) • Blue Mountain State: The Rise of Thadland            |  Movie            |  Comedies
...
 4) • Blue Planet II            |  TV Show            |  British TV Shows, Docuseries, Sci
ence & Nature TV...
 5) • Blue Ruin            |  Movie            |  Independent Movies, Thrillers...
 6) • Blue Streak            |  Movie            |  Action & Adventure, Comedies...
 7) • Blue Valentine            |  Movie            |  Dramas, Independent Movies, Romanti
c Movies...
 8) • BluffMaster!            |  Movie            |  Comedies, International Movies, Roman
tic Movies...
 9) • Blurred Lines: Inside the Art World            |  Movie            |  Documentaries.
..
10) • BNA            |  TV Show            |  Anime Series, International TV Shows...


Recommendations similar to: **Dune**

 1) • 1920            |  Movie            |  Horror Movies, International Movies, Thriller
s...
 2) • 3 Heroines            |  Movie            |  Dramas, International Movies, Sports Mo
vies...
 3) • Blue Mountain State: The Rise of Thadland            |  Movie            |  Comedies
...
 4) • Blue Planet II            |  TV Show            |  British TV Shows, Docuseries, Sci
ence & Nature TV...
 5) • Blue Ruin            |  Movie            |  Independent Movies, Thrillers...
 6) • Blue Streak            |  Movie            |  Action & Adventure, Comedies...
 7) • Blue Valentine            |  Movie            |  Dramas, Independent Movies, Romanti
c Movies...
 8) • BluffMaster!            |  Movie            |  Comedies, International Movies, Roman
tic Movies...
 9) • Blurred Lines: Inside the Art World            |  Movie            |  Documentaries.
..
10) • BNA            |  TV Show            |  Anime Series, International TV Shows...
 Found exact title → using full features


Recommendations similar to: **Stranger Things**

 1) • Beyond Stranger Things            |  TV Show            |  Stand-Up Comedy & Talk Sh
ows, TV Mysteries, TV Sci-Fi & Fantasy...
 2) • The Umbrella Academy            |  TV Show            |  TV Action & Adventure, TV My
steries, TV Sci-Fi & Fantasy...
 3) • The Messengers            |  TV Show            |  TV Dramas, TV Mysteries, TV Sci-Fi
& Fantasy...
 4) • Warrior Nun            |  TV Show            |  TV Action & Adventure, TV Mysteries,
TV Sci-Fi & Fantasy...
 5) • Sleepless Society: Nyctophobia            |  TV Show            |  International TV
Shows, TV Mysteries, TV Thrillers...
 6) • The OA            |  TV Show            |  TV Dramas, TV Mysteries, TV Sci-Fi & Fanta
sy...
 7) • The 4400            |  TV Show            |  TV Dramas, TV Mysteries, TV Sci-Fi & Fan
tasy...
 8) • Good Witch            |  TV Show            |  TV Dramas, TV Sci-Fi & Fantasy...
 9) • Reckoning            |  TV Show            |  Crime TV Shows, TV Dramas, TV Mysteries
...
10) • Haunted            |  TV Show            |  Reality TV, TV Horror, TV Thrillers...
 Found exact title → using full features


Recommendations similar to: **Article 15**
```

1) • Ginny Weds Sunny | Movie | Comedies, International Movies, Romantic Movies...
2) • No One Killed Jessica | Movie | Dramas, International Movies...
3) • Rukh | Movie | Dramas, Independent Movies, International Movies...
4) • Singh Saab the Great | Movie | Action & Adventure, Dramas, International Movies...
5) • Phantom | Movie | Action & Adventure, International Movies...
6) • Raja Natwarlal | Movie | Dramas, International Movies, Thrillers...
7) • Aiyaary | Movie | Action & Adventure, Dramas, International Movies...
8) • Raees | Movie | Action & Adventure, Dramas, International Movies...
9) • Har Kisse Ke Hisse: Kaamyaab | Movie | Dramas, International Movies...
10) • War Chhod Na Yaar | Movie | Comedies, Dramas, International Movies...

Recommendations similar to: **Dhurandhar**

1) • 1920 | Movie | Horror Movies, International Movies, Thrillers...
2) • 3 Heroines | Movie | Dramas, International Movies, Sports Movies...
3) • Blue Mountain State: The Rise of Thadland | Movie | Comedies...
4) • Blue Planet II | TV Show | British TV Shows, Docuseries, Science & Nature TV...
5) • Blue Ruin | Movie | Independent Movies, Thrillers...
6) • Blue Streak | Movie | Action & Adventure, Comedies...
7) • Blue Valentine | Movie | Dramas, Independent Movies, Romantic Movies...
8) • BluffMaster! | Movie | Comedies, International Movies, Romantic Movies...
9) • Blurred Lines: Inside the Art World | Movie | Documentaries...
10) • BNA | TV Show | Anime Series, International TV Shows...

Recommendations similar to: **Dhurandhar - The Revenge**

1) • Lady J | Movie | Dramas, Independent Movies, International Movies...
2) • Shuddhi | Movie | Dramas, International Movies, Thrillers...
3) • Uma Maheswara Ugra Roopasya | Movie | Comedies, Dramas, Independent Movies...
4) • Revenge | TV Show | Crime TV Shows, International TV Shows, TV Dramas...
5) • In Line | Movie | Dramas, International Movies...
6) • Cleopatra Jones | Movie | Action & Adventure, Classic Movies...
7) • Class of '83 | Movie | Dramas, International Movies, Thrillers...
8) • Needhi Singh | Movie | Action & Adventure, Dramas, International Movies...
9) • La Doña | TV Show | Spanish-Language TV Shows, TV Dramas...
10) • Monopoly (The Bank Of Luck) | Movie | Comedies, International Movies...
Found exact title → using full features

Recommendations similar to: **Dear Zindagi**

1) • Kapoor & Sons | Movie | Comedies, Dramas, International Movies...
2) • Chaahat | Movie | Dramas, International Movies, Romantic Movi

```
es...
 3) • Fitoor            |  Movie            |  Dramas, International Movies, Romantic Movi
es...
 4) • My Next Guest with David Letterman and Shah Rukh Khan            |  Movie
|  International Movies...
 5) • Chalte Chalte            |  Movie            |  Dramas, International Movies, Romant
ic Movies...
 6) • The Zoya Factor            |  Movie            |  Comedies, Dramas, International Mo
vies...
 7) • Shakti: The Power            |  Movie            |  Dramas, International Movies, Thr
illers...
 8) • Don 2            |  Movie            |  Action & Adventure, International Movies, Mu
sic & Musicals...
 9) • 2 States            |  Movie            |  Comedies, Dramas, International Movies...

10) • Asoka            |  Movie            |  Action & Adventure, Dramas, International Mo
vies...
```

In [ ]:

```

```

Q5] Can we forecast how many titles Netflix will add in 2022–2025?

In [142]:

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt

from sklearn.model_selection import train_test_split
from sklearn.metrics import r2_score, mean_absolute_error

yearly_data = df.groupby('year_added').size().reset_index(name='title_count')
yearly_data = yearly_data[yearly_data['year_added'] <= 2021]

X = yearly_data[['year_added']]
y = yearly_data['title_count']

# □ DO THIS ONLY ONCE
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

future_years = np.array([[2022], [2023], [2024], [2025]])
```

a) Linear Regression:

In [143]:

```
from sklearn.pipeline import Pipeline
from sklearn.preprocessing import StandardScaler, PolynomialFeatures
from sklearn.linear_model import LinearRegression

linear_pipeline = Pipeline([
    ('scaler', StandardScaler()),
    ('poly', PolynomialFeatures(degree=1)),
    ('model', LinearRegression())
])

linear_pipeline.fit(X_train, y_train)

y_pred_linear = linear_pipeline.predict(X_test)
linear_forecast = linear_pipeline.predict(future_years)
```

b) Adaboost:

In [144]:

```
from sklearn.ensemble import AdaBoostRegressor
from sklearn.tree import DecisionTreeRegressor

ada_pipeline = Pipeline([
    ('model', AdaBoostRegressor(
        estimator=DecisionTreeRegressor(max_depth=3),
        n_estimators=100,
        learning_rate=0.5,
        random_state=42
    ))
])

ada_pipeline.fit(X_train, y_train)

y_pred_ada = ada_pipeline.predict(X_test)
ada_forecast = ada_pipeline.predict(future_years)
```

c) Decision tree:

In [145]:

```
from sklearn.tree import DecisionTreeRegressor

tree_pipeline = Pipeline([
    ('model', DecisionTreeRegressor(
        max_depth=3,
        random_state=42
    ))
])

tree_pipeline.fit(X_train, y_train)

y_pred_tree = tree_pipeline.predict(X_test)
tree_forecast = tree_pipeline.predict(future_years)
```

In [146]:

```
forecast_df = pd.DataFrame({
    'Year': [2022, 2023, 2024, 2025],
    'Linear': linear_forecast.astype(int),
    'AdaBoost': ada_forecast.astype(int),
    'Decision Tree': tree_forecast.astype(int)
})

print(forecast_df)


print("\n\n\n")

results = pd.DataFrame({
    'Model': ['Linear', 'AdaBoost', 'Decision Tree'],
    'R2 Score': [
        r2_score(y_test, y_pred_linear),
        r2_score(y_test, y_pred_ada),
        r2_score(y_test, y_pred_tree)
    ],
    'MAE': [
        mean_absolute_error(y_test, y_pred_linear),
        mean_absolute_error(y_test, y_pred_ada),
        mean_absolute_error(y_test, y_pred_tree)
    ]
})

print(results)
```

```
   Year  Linear  AdaBoost  Decision Tree
0  2022    1283       117            117
1  2023    1400       117            117
2  2024    1517       117            117
```

```
3  2025    1634       117            117
```

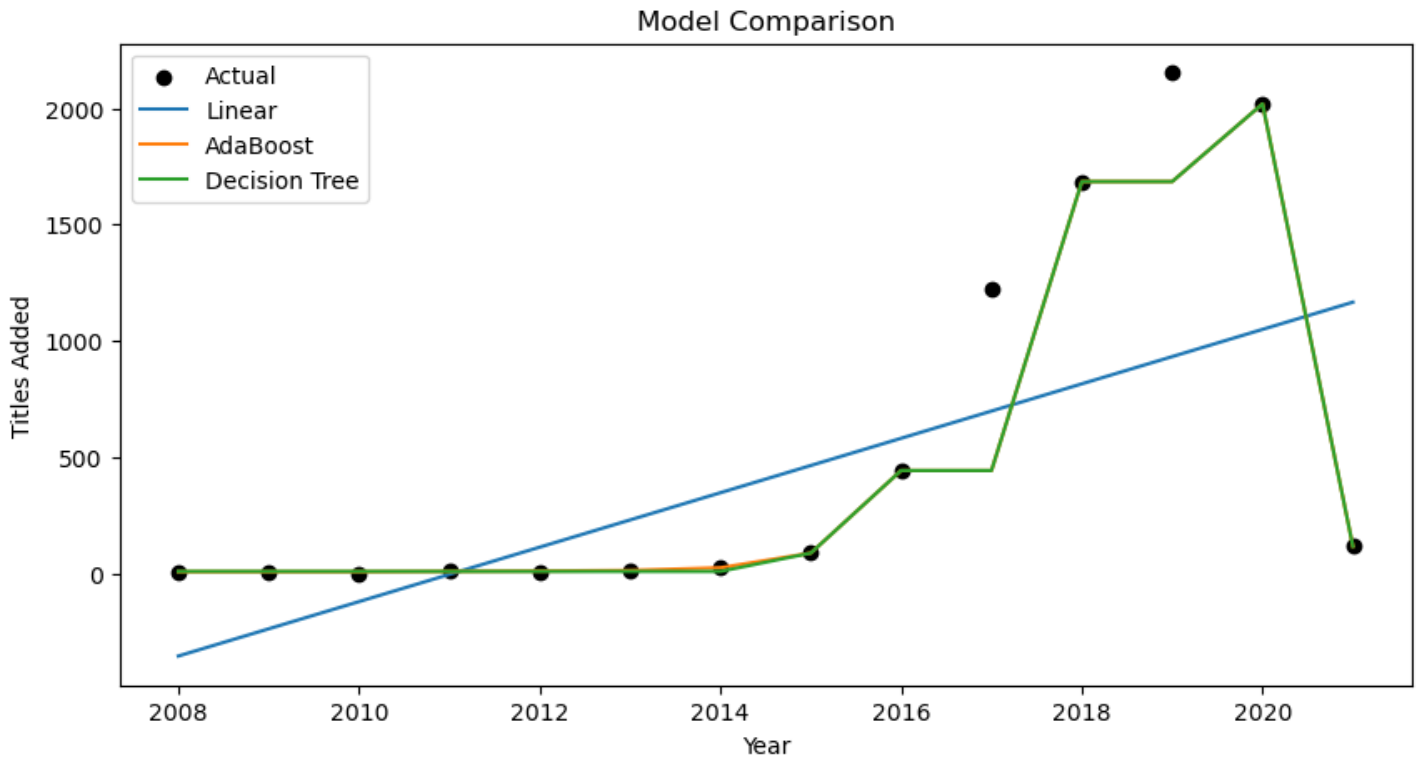
```
           Model  R2 Score         MAE
0         Linear  0.186415  701.208477
1       AdaBoost  0.643207  418.583333
2  Decision Tree  0.643199  419.055556
```

CONCLUSION] Tree-based models such as DecisionTreeRegressor and AdaBoostRegressor were tested but produced similar and unrealistic forecasts. This is because tree-based models do not extrapolate beyond the training data and tend to create step-like predictions, making them unsuitable for time-series forecasting tasks with limited features.

In [ ]:

In [147]:

```
# Sort for smooth plotting
sorted_idx = X['year_added'].argsort()
X_sorted = X.iloc[sorted_idx]

plt.figure(figsize=(10,5))

plt.scatter(X, y, color='black', label='Actual')

plt.plot(X_sorted, linear_pipeline.predict(X_sorted), label='Linear')
plt.plot(X_sorted, ada_pipeline.predict(X_sorted), label='AdaBoost')
plt.plot(X_sorted, tree_pipeline.predict(X_sorted), label='Decision Tree')

plt.legend()
plt.title('Model Comparison')
plt.xlabel('Year')
plt.ylabel('Titles Added')

plt.show()
```



In [ ]:

CONCLUSION] Linear regression performed best for trend modeling, while tree-based models failed to extrapolate future values due to their step-wise nature.

In [ ]:

In [ ]:

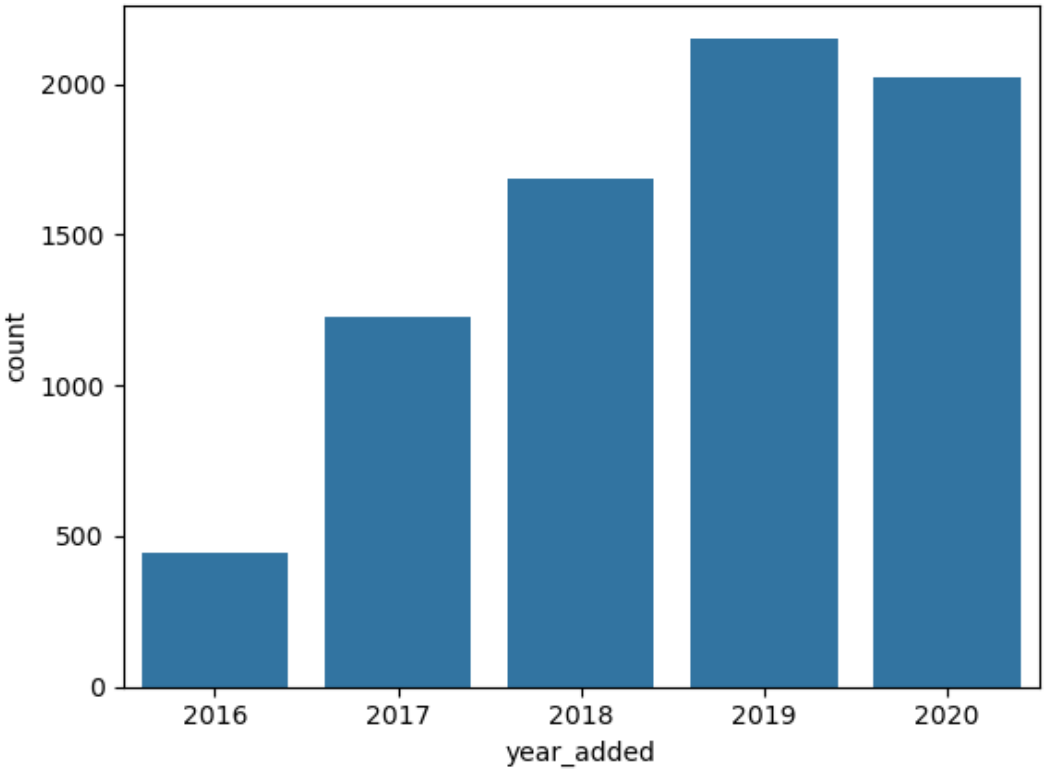
In [ ]:

In [ ]:

Q6] How is Netflix content growing over time?

In [148]:

```
data = df["year_added"].value_counts().reset_index()
data = data.head(5)
sns.barplot(x=data["year_added"],y=data["count"])
plt.show()
```



In [ ]:

In [149]:

```
yearly_data.sort_values(by="title_count",ascending=False).head(5)
```

Out[149]:

	year_added	title_count

	year_added	title_count
11	2019	2153
12	2020	2019
10	2018	1685
9	2017	1225
8	2016	443

CONCLUSION] Netflix content has shown a consistent upward trend over the years, with significant growth in recent years. Linear models were able to capture this trend effectively, while tree-based models failed due to their inability to extrapolate beyond training data. This highlights the importance of selecting models based on problem type rather than complexity.

In [ ]:

```

```

In [ ]:

```

```

Q7] Top 5 directors:

In [150]:

```
data = df["director"].value_counts().reset_index().head()
data
```

Out[150]:

	director	count
0	Unknown	2389
1	Raúl Campos, Jan Suter	18
2	Marcus Raboy	16
3	Jay Karas	14
4	Cathy Garcia-Molina	13

Q8] Top 5 countries:

In [151]:

```
data = df["first_country"].value_counts().reset_index().head(5)
data
```

Out[151]:

	first_country	count
0	United States	2883
1	India	956
2	United Kingdom	577
3	Unknown	507
4	Canada	259

In [ ]:

```

```

In [ ]:

```

```